

Mastering Neural Networks and Model Regularization

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- Module 1
- Course Introduction
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Course Overview

Reading
- ✓

Instructor Biography - Dr. Erhan Guven

Reading

- Module 2
- Multilayer Artificial Neural Networks
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- ✓

Multilayer Artificial Neural Networks Overview

Video
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Neural Networks

Video
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Reading References

Reading
- ✓

Overview of Multilayer Perceptrons

Practice Assignment
- ✓

Forward Propagation Algorithm

Video
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Neural Network Design and Implementation

Video
- ✓

Reading References

Reading
- ✓

Forward Propagation

Practice Assignment
- ✓

Graded Assessment

Graded Assignment • Grade: 90%
- ✓

Practice Lab: Mining Patterns in Alice in Wonderland & Building a Neural Network on MNIST Dataset

Lab

- Module 3
- Model Regularization
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Introduction to Regularization Overview

Video
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Regularization and Neural Network Dropout

Video
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Reading References

Reading
- ✓

Implementation of Dropout in Neural Networks

Practice Assignment
- ✓

Regularization Demonstration

Video
- ✓

Reading References

Reading
- ✓

Practical Application of Regularization Techniques in ML

Practice Assignment
- ✓

Self-Reflective Reading: Designing Neural Networks

Reading
- ✓

Graded Assessment

Graded Assignment • Grade: 100%
- ✓

Practice Lab: Cyber Intrusion Detection Systems

Lab

- Module 4
- PyTorch
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PyTorch Overview

Video
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Introduction to PyTorch

Video
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Reading References

Reading
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Getting Started with PyTorch

Practice Assignment
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PyTorch Demonstration

Video
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Reading References

Reading
- ✓

Hands-on Application of PyTorch in Deep Learning

Practice Assignment
- ✓

Graded Assessment

Graded Assignment • Grade: 90%
- 🔒

Hands-on lab - Implementing Fraud Detection Models with PyTorch and Scikit-Learn

Lab

- Module 5
- Convolutional Neural Networks
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Convolutional Neural Networks

Video
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Reading References

Reading
- ✓

Exploring Convolutional Neural Networks (CNNs)

Practice Assignment
- ✓

Audio Signal Classification

Video
- ✓

Reading References

Reading
- ✓

Analyzing Audio Time Signals with Spectrograms

Practice Assignment
- ✓

Self-Reflective Reading: CNNs and RNNs

Reading
- ✓

Graded Assessment

Graded Assignment • Grade: 90%
- ✓

Graded Lab: Image Classification Using PyTorch

Programming Assignment • Grade: 100%

Practice Lab: Cyber Intrusion Detection Systems

Launch lab

Instructions

To get started, click the "**Launch Lab**" button. This will open the lab environment in a new browser tab, allowing you to interact with the virtual lab setup provided for this course.

If you prefer to work offline, you can download the lab notebook to your local environment by clicking on the file below.

 Hands-on Lab - Cyber_Intrusion Detection Systems

IPYNB File

The lab notebook (.ipynb file) includes important instructions, questions, and coding tasks to guide you through the lab.

Datasets utilized in this lab:

 Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX

CSV File

 Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX

CSV File

 Tuesday-WorkingHours.pcap_ISCX

CSV File

 Wednesday-workingHours.pcap_ISCX

CSV File

Datasets for Practice:

 Friday-WorkingHours-Morning.pcap_ISCX

CSV File

 Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX

CSV File

 Friday-WorkingHours-Afternoon-DDos.pcap_ISCX

CSV File

Be sure to follow the directions in the notebook closely to make the most of your learning experience. Enjoy exploring and experimenting in the virtual lab!

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